Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITPPA1-44

Group: 4

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

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Signature	Date
KBatista	08/11/2024

Lecturer's Comments:

Marks Awarded:	**%**

Signature	Date

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Section A

Section A

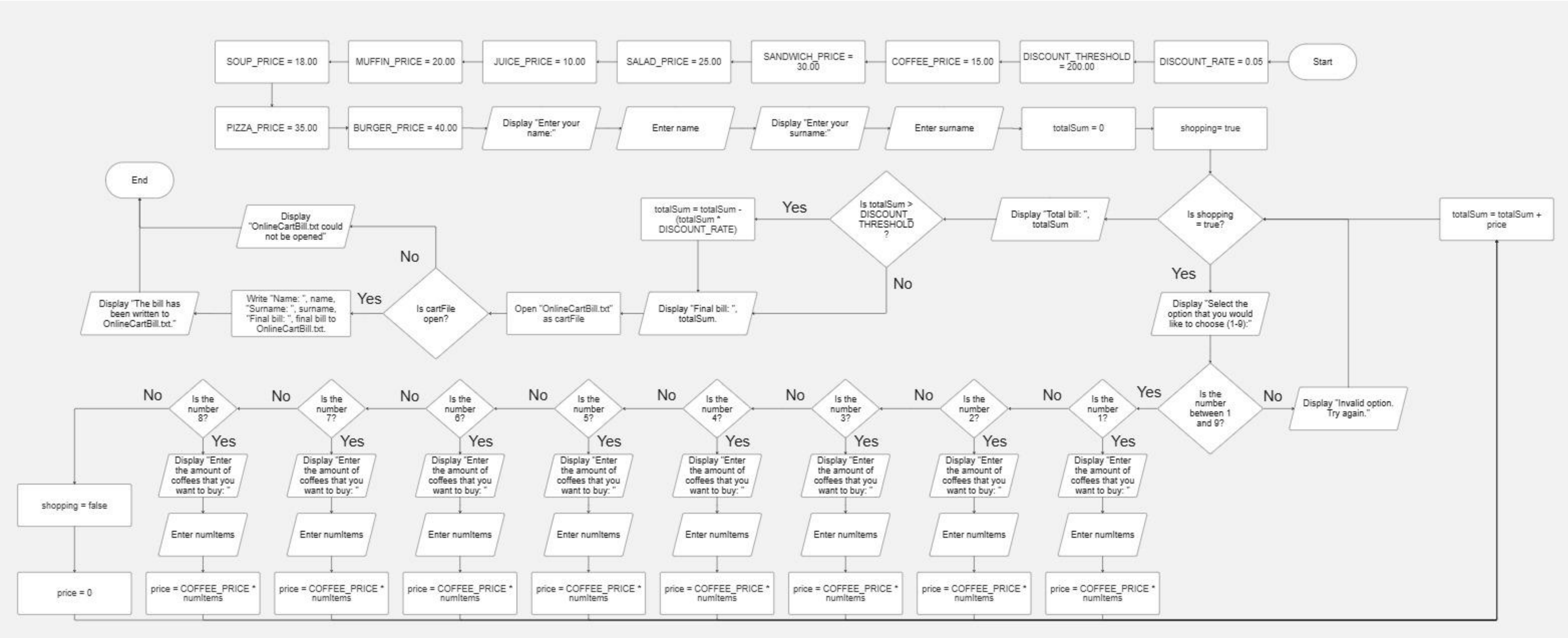
Question 1

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
	//Constants and variables are declared
	const double DISCOUNT_RATE = 0.10;
	const double DISCOUNT_THRESHOLD = 100.00;

	const double COFFEE_PRICE = 15.00;
	const double SANDWICH_PRICE = 30.00;
	const double SALAD_PRICE = 25.00;
	const double JUICE_PRICE = 10.00;
	const double MUFFIN_PRICE = 20.00;
	const double PIZZA_PRICE = 35.00;
	const double SOUP_PRICE = 18.00;
	const double BURGER_PRICE = 40.00;

	string name, surname;
	int item, numItems;
	double totalSum, finalSum, price;
	ofstream cafeteriaFile;

	//Information is gathered from the user
	cout << "Enter your name please: ";
	cin >> name;
	cout << endl;
	cout << "Enter your surname please: ";
	cin >> surname;
	cout << endl;
	cout << "How many items would you like to order (up to a maximum of 8)?: ";
	cin >> numItems;
	cout << endl << endl;

	cout << "Cafeteria Menu:\n";
	cout << "1. Coffee - " << COFFEE_PRICE << endl;
	cout << "2. Sandwich - " << SANDWICH_PRICE << endl;
	cout << "3. Salad - " << SALAD_PRICE << endl;
	cout << "4. Juice - " << JUICE_PRICE << endl;
	cout << "5. Muffin - " << MUFFIN_PRICE << endl;
	cout << "6. Pizza slice - " << PIZZA_PRICE << endl;
	cout << "7. Soup - " << SOUP_PRICE << endl;
	cout << "8. Burger - " << BURGER_PRICE << endl;

	totalSum = 0;
	price = 0;
	//The user picks the item that they would like to buy and its price is added to the total
	for (int i = 1; i <= numItems; i++) {
		cout << "\nSelect what order " << i << " should be (1-8):";
		cin >> item;

		switch (item) {
		case 1:
			price = COFFEE_PRICE;
			break;
		case 2:
			price = SANDWICH_PRICE;
```

```
                break;
            case 3:
                price = SALAD_PRICE;
                break;
            case 4:
                price = JUICE_PRICE;
                break;
            case 5:
                price = MUFFIN_PRICE;
                break;
            case 6:
                price = PIZZA_PRICE;
                break;
            case 7:
                price = SOUP_PRICE;
                break;
            case 8:
                price = BURGER_PRICE;
                break;
            }

            totalSum = totalSum + price;
        }

        //A discount is applied to the final sum if the total is more than R100
        cout << "\n\nTotal Bill: R" << totalSum;
        if (totalSum > DISCOUNT_THRESHOLD) {
            finalSum = totalSum - (totalSum * DISCOUNT_RATE);
            cout << "\nA discount was applied.\n";
        }
        else {
            finalSum = totalSum;
            cout << "\nNo discount was applied.\n";
        }

        cout << "\nFinal Bill: R" << finalSum << endl;

        //User information is sent to the file, CafeteriaBill.txt
        cafeteriaFile.open("CafeteriaBill.txt", ios::trunc); (prakhargvp, 2024)
        if (cafeteriaFile.is_open()) {
            cafeteriaFile << "Name: " << name << endl;
            cafeteriaFile << "Surname: " << surname << endl;
            cafeteriaFile << "Final bill: R" << finalSum << endl;
            cafeteriaFile.close();
            cout << "The bill has been written to CafeteriaBill.txt.\n";
        }
        else {
            cout << "CafeteriaBill.txt could not be opened.\n";
        }
}
```

Question 2

2.1)



2.2)

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
	//Constants and variables are declared
	const double DISCOUNT_RATE = 0.05;
	const double DISCOUNT_THRESHOLD = 200.00;

	const double COFFEE_PRICE = 15.00;
	const double SANDWICH_PRICE = 30.00;
	const double SALAD_PRICE = 25.00;
	const double JUICE_PRICE = 10.00;
	const double MUFFIN_PRICE = 20.00;
	const double PIZZA_PRICE = 35.00;
	const double SOUP_PRICE = 18.00;
	const double BURGER_PRICE = 40.00;

	string name, surname;
	int option, numItems;
	double totalSum, finalSum, price;
	bool shopping = true;
	ofstream cartFile;

	//Information is gathered from the user
	cout << "Enter your name please: ";
	cin >> name;
	cout << endl;
	cout << "Enter your surname please: ";
	cin >> surname;
	cout << endl;

	cout << "Online Shopping Menu:\n\n";
	cout << "1. Coffee - " << COFFEE_PRICE << endl;
	cout << "2. Sandwich - " << SANDWICH_PRICE << endl;
	cout << "3. Salad - " << SALAD_PRICE << endl;
	cout << "4. Juice - " << JUICE_PRICE << endl;
	cout << "5. Muffin - " << MUFFIN_PRICE << endl;
	cout << "6. Pizza slice - " << PIZZA_PRICE << endl;
	cout << "7. Soup - " << SOUP_PRICE << endl;
	cout << "8. Burger - " << BURGER_PRICE << endl << endl;
	cout << "9. Go to cart" << endl;

	totalSum = 0;
	//All purchase options will keep being asked until the user chooses option 9 to end the loop
	and calculate the final bill
	while (shopping == true) {
		cout << "\nSelect the option that you would like to choose (1-9):";
		cin >> option;

		switch (option) {
		case 1:
			cout << "\nEnter the amount of coffees that you want to buy: ";
			cin >> numItems;
			price = COFFEE_PRICE * numItems;
			break;
		case 2:
			cout << "\nEnter the amount of sandwiches that you want to buy: ";
			cin >> numItems;
			price = SANDWICH_PRICE * numItems;
			break;
		case 3:
			cout << "\nEnter the amount of salads that you want to buy: ";
			cin >> numItems;
			price = SALAD_PRICE * numItems;
```

```
                break;
            case 4:
                cout << "\nEnter the amount of juices that you want to buy: ";
                cin >> numItems;
                price = JUICE_PRICE * numItems;
                break;
            case 5:
                cout << "\nEnter the amount of muffins that you want to buy: ";
                cin >> numItems;
                price = MUFFIN_PRICE * numItems;
                break;
            case 6:
                cout << "\nEnter the amount of pizza slices that you want to buy: ";
                cin >> numItems;
                price = PIZZA_PRICE * numItems;
                break;
            case 7:
                cout << "\nEnter the amount of soups that you want to buy: ";
                cin >> numItems;
                price = SOUP_PRICE * numItems;
                break;
            case 8:
                cout << "\nEnter the amount of burgers that you want to buy: ";
                cin >> numItems;
                price = BURGER_PRICE * numItems;
                break;
            case 9:
                shopping = false;
                price = 0;
                break;
            default:
                cout << "\nInvalid option. Try again please.";
                price = 0;
                break;
            }

            //The total is calculated
            totalSum = totalSum + price;
        }

        //A discount is applied to the final sum if the total is more than R200
        cout << "\n\nTotal bill: R" << totalSum;
        if (totalSum > DISCOUNT_THRESHOLD) {
            finalSum = totalSum - (totalSum * DISCOUNT_RATE);
            cout << "\nA discount was applied.\n";
        }
        else {
            finalSum = totalSum;
            cout << "\nNo discount was applied.\n";
        }
        cout << "\nFinal bill: R" << finalSum << endl;

        //User information is sent to the file, OnlineCartBill.txt
        cartFile.open("OnlineCartBill.txt", ios::trunc); (prakhargvp, 2024)
        if (cartFile.is_open()) {
            cartFile << "Name: " << name << endl;
            cartFile << "Surname: " << surname << endl;
            cartFile << "Final bill: R" << finalSum << endl;
            cartFile.close();
            cout << "The bill has been written to OnlineCartBill.txt.\n";
        }
        else {
            cout << "OnlineCartBill.txt could not be opened.\n";
        }
}
```

Question 3

```
#include <iostream>
using namespace std;

int main() {
	int scores[5], highestScore, lowestScore; //Variable declaration
	double totalScores, averageScore;

	totalScores = 0;
	highestScore = 0;
	lowestScore = 100;

	//Input the scores and determine the highest as well as the lowest score that were inputted
	for (int i = 0; i <= 4; i++) {
		cout << "Enter student " << (i + 1) << " score: ";
		cin >> scores[i];
		cout << endl;

		if (highestScore < scores[i]) {
			highestScore = scores[i];
		}
		if (lowestScore > scores[i]) {
			lowestScore = scores[i];
		}
		totalScores = totalScores + scores[i];
	}

	cout << "Scores entered:\n";
	//Displays the scores in order
	for (int i = 0; i <= 4; i++) {
		cout << "Student " << (i + 1) << ": " << scores[i] << endl;
	}
	cout << endl;

	//Calculate and display the average score
	averageScore = ((totalScores / 500) * 100);
	cout << "Average score: " << (averageScore) << endl;

	cout << "Highest score: " << (highestScore) << endl;
	cout << "Lowest score: " << (lowestScore) << endl;

	return 0;
}
```

Question 4

```
#include <iostream>
#include <sstream>
#include <locale>
using namespace std;

bool createAccount(string &name, int &accountNumber, double &balance) {
	//Variable declaration
	string userName, userNumber, strDeposit;
	bool validAccount = false, validDeposit = false;
	int validNum = 0, decimalsFound = 0;
	double initialDeposit;

	cout << "Enter your name: ";
	getline(cin >> ws, userName);
	cout << endl;

	cout << "Enter your account number (it is 5 numbers in length): ";
	getline(cin >> ws, userNumber);
	cout << endl;

	//Determine if the account number doesn't have any letters and is five characters long
	if (userNumber.length() == 5) {
		for (int i = 0; i <= userNumber.length(); i++) {
			if ((userNumber[i] >= '0' && userNumber[i] <= '9')) {
				validNum++;
			}
		}
		if (validNum == userNumber.length()) {
			validAccount = true;
		}
	}

	if (validAccount == true) {
		cout << "Enter your initial deposit (it must be more than R0.00): R";
		getline(cin >> ws, strDeposit);
		cout << endl;

		validNum = 0;
		//Determine if the deposit value doesn't have any letters and an incorrect amount of
		decimal points
		for (int i = 0; i <= strDeposit.length(); i++) { (IT, 2022)
			if ((strDeposit[i] >= '0' && strDeposit[i] <= '9') || (strDeposit[i] == '.')) {
				validNum++;
				if (strDeposit[i] == '.') {
					decimalsFound++;
				}
			}
		}

		//If each character in the deposit value is correct and the decimal points found are
		less than 2, change the deposit value from a string to a double
		if (validNum == strDeposit.length() && decimalsFound < 2) {
			initialDeposit = stod(strDeposit); (ryan, 2024)
			//Determine if the deposit is more than 0
			if (initialDeposit > 0) {
				validDeposit = true;
			}
		}

		//If all the required information is valid, then create the account
		if (validDeposit == true) {
```

```
                // If all the required information is valid and the deposit will not create an
                overflow, create the account
                if (initialDeposit < numeric_limits<unsigned int>::max()) { (satwiksuman, 2024)
                        cout << "Account created successfully!\n";
                        name = userName;
                        istringstream(userNumber) >> accountNumber; (Kulyyev, 2018)
                        balance = initialDeposit;
                        return(true);
                }
                else {
                        cout << "Initial deposit is too big, it will create an overflow.\n";
                }
            }
            else {
                cout << "Invalid initial deposit.\n";
            }
        }
        else {
            cout << "Invalid account number.\n";
        }
}

void depositMoney(double &balance) {
        //Variable declaration
        double deposit, balanceTest = balance;
        string strDeposit;
        int validNum = 0, decimalsFound = 0;
        bool validDeposit = false;

        cout << "Enter your deposit: R";
        getline(cin >> ws, strDeposit);
        cout << endl;

        //Determine if the deposit value doesn't have any letters and an incorrect amount of decimal
        points
        for (int i = 0; i <= strDeposit.length(); i++) { (IT, 2022)
            if ((strDeposit[i] >= '0' && strDeposit[i] <= '9') || (strDeposit[i] == '.')) {
                validNum++;
                if (strDeposit[i] == '.') {
                        decimalsFound++;
                }
            }
        }

        //If each character in the deposit value is correct and the decimal points found are less than
        2, change the deposit value from a string to a double
        if (validNum == strDeposit.length() && decimalsFound < 2) {
            deposit = stod(strDeposit); (ryan, 2024)
            //Determine if the deposit is more than 0
            if (deposit > 0) {
                validDeposit = true;
            }
        }

        if (validDeposit == true) {
            // If the deposit value is valid and an overflow will not happen, process the deposit
            balanceTest = balance + deposit;
            if (balanceTest < numeric_limits<unsigned int>::max()) { (satwiksuman, 2024)
                balance = balance + deposit;
                cout << "Deposit completed!\n";
            }
            else {
                cout << "The deposit is too big, it will create an overflow.\n";
            }
        }
        else {
            cout << "Invalid deposit.\n";
```

```
	}
}

void withdrawMoney(double &balance) {
	//Variable declaration
	double withdraw;
	string strWithdraw;
	int validNum = 0, decimalsFound = 0;
	bool validWithdraw = false;

	cout << "Enter your withdraw: R";
	getline(cin >> ws, strWithdraw);
	cout << endl;

	//Determine if the withdraw value doesn't have any letters and an incorrect amount of decimal
	points
	for (int i = 0; i <= strWithdraw.length(); i++) { (IT, 2022)
		if ((strWithdraw[i] >= '0' && strWithdraw[i] <= '9') || (strWithdraw[i] == '.')) {
			validNum++;
			if (strWithdraw[i] == '.') {
				decimalsFound++;
			}
		}
	}

	//If each character in the withdraw value is correct and the decimal points found are less than
	2, change the withdraw value from a string to a double
	if (validNum == strWithdraw.length() && decimalsFound < 2) {
		withdraw = stod(strWithdraw); (ryan, 2024)
		//Determine if the withdraw is less than the balance
		if (withdraw < balance) {
			validWithdraw = true;
		}
	}

	//If the withdraw value is valid, then process the withdraw
	if (validWithdraw == true) {
		balance = balance - withdraw;
		cout << "Withdraw completed!\n";
	}
	else {
		cout << "Invalid withdraw.\n";
	}
}

void checkBalance(const double &balance) {
	//Display the balance
	cout << "Current balance: R" << balance << endl;
}

void displayAccountDetails(const string &name, const int &accountNumber, const double &balance) {
	//Display all account details
	cout << "Account holder: " << name << endl;
	cout << "Account number: " << accountNumber << endl;
	cout << "Current balance: R" << balance << endl;
}
int menu() {
	//Variable declaration
	int choice;

	//Display the menu and receive input from the user
	cout << "--- Bank Account Management System ---\n";

	cout << "1. Create account\n";
	cout << "2. Deposit money\n";
	cout << "3. Withdraw money\n";
	cout << "4. Check balance\n";
```

```cpp
	cout << "5. Display account details\n";
	cout << "6. Exit\n\n";
	cout << "Enter your choice (1-6):";
	cin >> choice;
	return choice;
}
int main() {
	//Variable declaration
	string name;
	int accountNumber, choice;
	double balance;
	bool exitSystem = false, accountCreated = false;
	//Loops through the options until the user chooses option 6 that ends the loop and exits the
program
	while (exitSystem == false) {
		choice = menu();

		switch (choice) {
		case 1: //Create an account
			accountCreated = createAccount(name, accountNumber, balance);
			break;
		case 2: //Deposit money into the account
			if (accountCreated == true) {
				depositMoney(balance);
			}
			else {
				cout << "An account must first be created!\n";
			}
			break;
		case 3: //Withdraw money from the account
			if (accountCreated == true) {
				withdrawMoney(balance);
			}
			else {
				cout << "An account must first be created!\n";
			}
			break;
		case 4: //Display the balance
			if (accountCreated == true) {
				checkBalance(balance);
			}
			else {
				cout << "An account must first be created!\n";
			}
			break;
		case 5: //Display all account details
			if (accountCreated == true) {
				displayAccountDetails(name, accountNumber, balance);
			}
			else {
				cout << "An account must first be created!\n";
			}
			break;
		case 6: //Exit program
			exitSystem = true;
			cout << "Exiting the system. Goodbye!";
			break;
		default:
			cout << "Invalid choice. Try again!";
			break;
		}
		cout << endl;
	}
	return 0;
}
```

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